

Computer Vision-Assisted Semantic Material Identification for High-Fidelity Digital Twins

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Abstract

Ray-tracing-based digital twins critically depend on accurate electromagnetic material modeling, yet reconstructed meshes are often imported with simplified or default material assignments, degrading radio propagation predictions. This paper proposes a computer vision-assisted semantic material identification pipeline that converts a reconstructed 3D mesh into a high-fidelity scene for Sionna RT. Simulation results show that the proposed pipeline yields measurable, position-dependent channel differences, supporting 5G/6G digital twin construction.

I. Introduction

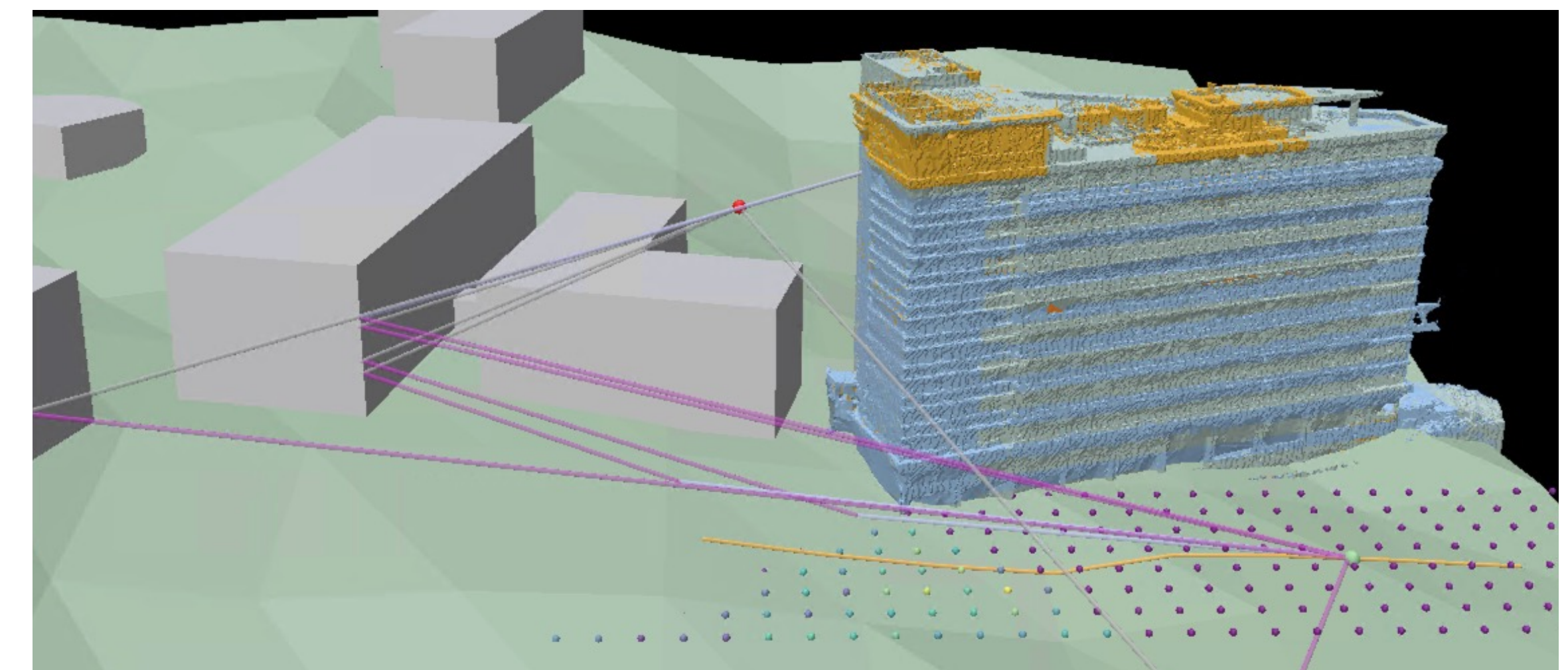
❖ Why material fidelity matters

FTC: Fusion Technology Center

- Wireless digital twins enable 5G/6G optimization via RF propagation simulation (e.g., Sionna RT); accuracy depends on mesh material modeling, especially at high frequencies
- Reconstructed meshes are often simplified to a single homogeneous material (e.g., concrete), degrading propagation predictions

❖ Objective

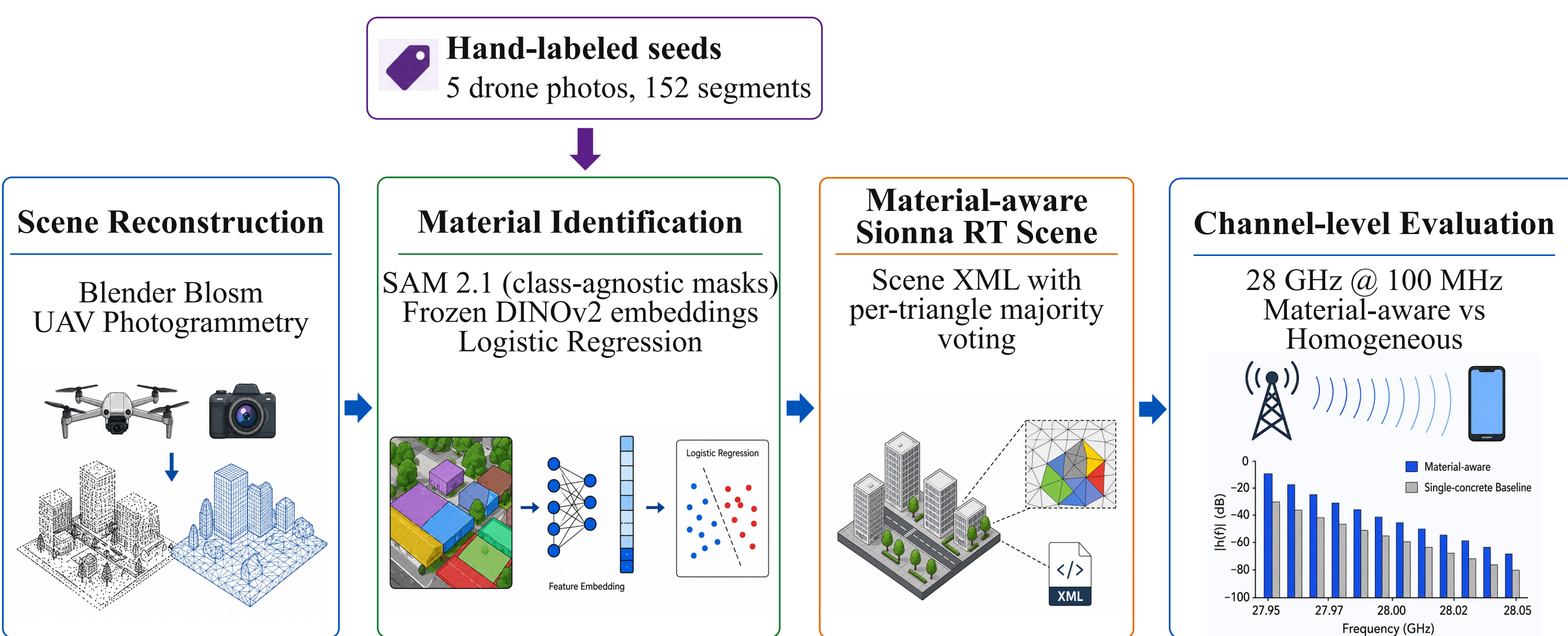
- A CV-assisted pipeline combining SAM 2.1 and DINOv2 to build a material-aware Sionna RT scene of the FTC building, evaluated against a single-concrete baseline at 28 GHz



< 28GHz propagation analysis with the Sionna RT GUI >

II. Proposed Pipeline

❖ CV-assisted material identification pipeline



❖ Pipeline details

- Scene: terrain / buildings (Blender Blosm) + FTC mesh (UAV photogrammetry, DJI Mini 5 Pro)
- Labels: 5 UAV photos → 152 segments, DINOv2 embeddings + logistic regression (5-fold accuracy 0.848)
- Masks: SAM 2.1 on remaining images → per-triangle majority voting → Sionna RT XML
- Mapping: walls/roofs → ITU concrete, windows → glass, frames → metal, terrain → 28 GHz ground ($\epsilon_r = 3.0$, $\sigma = 0.05$ S/m), unclassified → concrete

III. Channel-Level Evaluation Framework

❖ CFR-based metrics quantify material-mapping impact

- Metrics

- NMSE (material-aware vs single-concrete)

$$NMSE = \frac{\sum |h_{mat} - h_{base}|^2}{\sum |h_{mat}|^2} \quad NMSE \rightarrow -\infty \text{ dB for identical CFRs}$$

- Cosine similarity (structural alignment)

$$\rho = \frac{|h_{mat}^H h_{base}|}{\|h_{mat}\| \cdot \|h_{base}\|} \quad \rho \approx 1 \text{ for near-parallel CFRs}$$

- Per-position RSS error (signed, dB)

$$\Delta P_{RSS} = 10 \log_{10} |h_{mat}|^2 - 10 \log_{10} |h_{base}|^2 \text{ [dB]}$$

- Throughput proxy (Shannon-style)

$$T(h) = B \cdot \mathbb{E}_f [\log_2(1 + \gamma(f))]$$

$$\gamma(f) = P_{tx} |h(f)|^2 / N$$

$$N = -174 + 10 \log_{10}(B) + NF$$

h_{mat} : material-aware CFR

h_{base} : baseline CFR

B: bandwidth

P_{tx} : transmit power, 30 dBm

N: noise power, -87 dBm

NF: noise figure, 7 dB

❖ Remarks

- Signed P_{RSS} preserves the direction of mismatch
- Throughput proxy is a Shannon upper bound

IV. Experimental Setup and Results

❖ Setup

Parameter	Value
Frequency / Bandwidth	28 GHz / 100 MHz
TX	One array
UE Trajectory / Positions	Up to 50 m resampled / 50 positions
Max ray-tracing depth	6

❖ Canonical results (seed = 42)

m.s. = material-sensitive subset

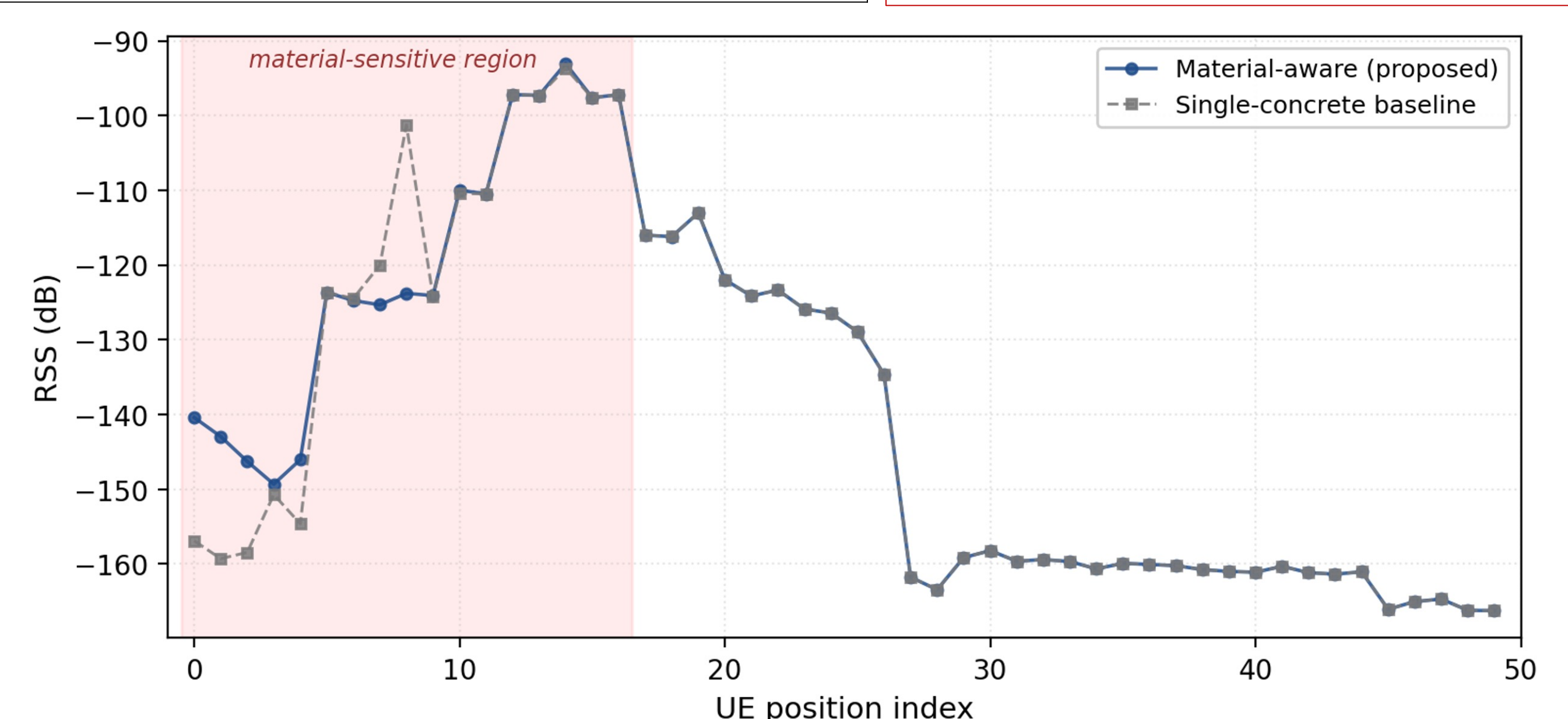
Metric	Material-aware	Single-concrete	Δ
Mean RSS, all (dB)	-139.94	-140.50	+0.56
Mean RSS, m.s. (dB)	-129.36	-130.36	+1.00
Mean T_{proxy} (Mbps)	89.22	99.49	-10.27
Mean T_{proxy} , m.s. (Mbps)	128.53	146.87	-18.34
Mean path count	12.4	12.8	-0.4

❖ Fair condition frequency comparison

Metric	3.5 GHz	28 GHz
Active Global NMSE (dB)	-19.53	-19.60
Active mean cos. sim.	0.444	0.430
Mean T_{proxy} error (Mbps)	74.9	13.8
Both active / base only	32 / 18	32 / 18

Seed	Global NMSE
42	-11.89 dB (canonical)
42-45	-25.0 ± 9.8 dB

* Both active / base only: # UE positions where both scenes yield an active channel



< Per-position RSS at 28 GHz: material-aware vs single-concrete baseline >

- 28/50 UE positions material-sensitive (per-position NMSE > -60 dB), seed-averaged 24.3 ± 2.6
- Throughput-proxy gap of 18 Mbps in the material-sensitive subset, up to 22 dB RSS gap at p = 8
- Fair-condition: -19.53 dB (3.5 GHz) vs -19.60 dB (28 GHz)
∴ position-dependent, not frequency-dependent

V. Conclusion

- Position-dependent channel differences vs a single-concrete baseline at 28 GHz → supports site-specific digital twins
- Future work: hand-label expansion, texture-aware platforms, alternative UE trajectories & TX/RX placements